

HackOn with Amazon

A Universe of Opportunity

48-Hour Hackathon | Solution Document

Team Information

Field	Details
Team Name	Jaguars
Hackathon Theme	PocketBuddy – AI Financial & Wellness Assistant for Students
Date	June 15, 2026

Team Members

Name	College / University	Role	Email
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1. Problem Statement & Relevance

The Problem

Millions of students struggle silently with a multidimensional crisis: the crushing overlap of budgeting issues, food expenses, emotional stress, irregular sleep, and the heavy burden of balancing academics with social life. Existing solutions focus on isolated silos—finance, fitness, or productivity—failing to grasp the interconnected reality of modern student living, where financial anxiety directly triggers academic burnout and sleep deprivation.

Why It Matters

When a student's budget is drained, their mental health immediately follows. Studies show that financial stress is a leading cause of college dropouts and severe academic decline. The cost of inaction is a generation paralyzed by debt and anxiety before they even enter the workforce. Traditional tools tell them *what* went wrong; they need a tool that helps

What Makes This Novel

What if students had an AI companion that actually understood their reality? PocketBuddy doesn't just show pie charts; it uses **Dual-Vector AI Analysis**. It correlates spending habits with emotional states—recognizing, for example, that a spike in fast-food spending correlates with high stress and irregular sleep. It preemptively intervenes, recommending affordable travel, cheap local food, detecting burnout patterns, and encouraging healthy routines before the student hits a breaking point.

2. Customer & Solution

Target Customer

The **"Overwhelmed Scholar" (Ages 18-25)**: A college student balancing classes, a tight budget, and social pressures. They want to manage their monthly expenses and find affordable food options, but feel too stressed to manage complex spreadsheets. They need an empathetic, automated companion.

How We Solve It

PocketBuddy is a unified dashboard that acts as a 24/7 personalized support system for both finance and wellness.

- **Dual-Tracking Engine**: Manages monthly expenses while simultaneously logging daily habits (sleep, study hours, mood).
- **Context-Aware AI Coach**: Detects burnout patterns and provides personalized

User Workflow

1. **Onboarding:** User sets initial financial boundaries and wellness baselines (sleep goals, stress triggers).
2. **Daily Interaction:** The dashboard provides a unified "Health & Wealth" score. The user logs their mood in 5 seconds.
3. **AI Intervention:** If spending spikes during a low-sleep week, the AI Coach pings the user with empathetic advice, a cheap local food recommendation, and a nudge to prioritize rest.

Working Prototype

Live Demo: <https://budgety-bud.onrender.com>

Code Repository: <https://github.com/Ashutosh-kumar-06/Budgety-bud.git>

3. Tech Architecture & Scaling

Tech Stack

Layer	Technology	Why
Frontend	React 19, Vite, TypeScript	Lightning-fast HMR, strict type safety for financial data, modern component architecture.
Backend	Express.js, TypeScript	High-performance, unopinionated routing with robust middleware for JWT auth.
Data/ML	MongoDB Atlas, Gemini AI	Flexible schema for complex user profiles; Gemini provides fast, nuanced natural language processing for the Coach.
Infra	Render, AWS (Planned)	PaaS for rapid deployment; planned migration to ECS/Ec2 for infinite scaling

Key Algorithms & Complexity

Dynamic Correlation Engine ($O(N \log N)$): The system aggregates transaction vectors (timestamp, amount, category) against wellness vectors (timestamp, mood score, sleep hours). By employing time-series smoothing, it identifies statistically significant overlaps between financial strain and emotional dips, feeding this context window directly to the Gemini LLM prompt to trigger burnout detection alerts.

Scaling Strategy

Currently deployed horizontally on Render. For 1000x growth:

- **Compute:** Migrate the Express backend to AWS ECS (Fargate) with auto-scaling groups based on CPU utilization.
- **Database:** Shard MongoDB Atlas by `university_id` to localize queries, as student interactions (deals, resources) are highly geographic.

4. Future Vision

Where This Goes

Within 3 years, PocketBuddy will become the **default operating system for student life**, integrated directly with University financial aid offices and campus health centers to provide a real-time safety net for at-risk students.

Roadmap

Horizon	Milestone	Impact
0-3 mo	Plaid API integration for live bank syncing.	5,000 active users; automated data ingestion.
3-6 mo	"Campus Connect" B2B portal for universities.	50,000 users; universities can monitor anonymized campus stress/financial health.

Multi-Segment Expansion

While starting with higher education, the core "Finance + Wellness" engine is highly transferable.

- **Gig Economy Workers:** Freelancers dealing with variable income stress.
- **Young Professionals:** Navigating first salaries, 401ks, and corporate burnout.

Value Impact

By reducing financial-stress-related college dropout rates by just 1%, PocketBuddy preserves millions in tuition revenue for universities and protects billions in lifetime earning potential for students, breaking cycles of debt and depression.

Thank You

<https://budgety-bud.onrender.com>